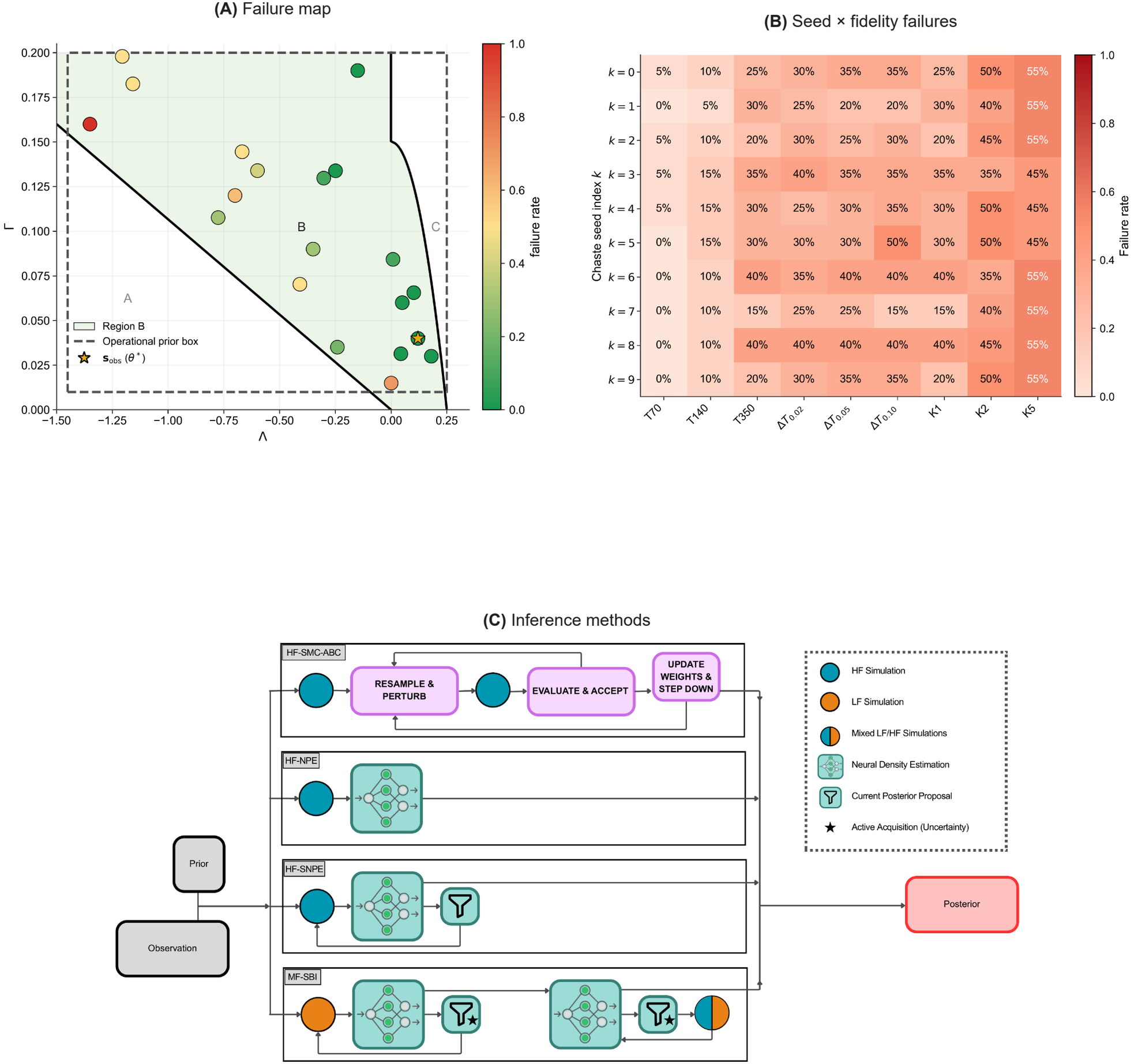

(A) Failure map
Γ
Λ
A
B
C
Region B
Operational prior box
s_obs (θ*)
failure rate
(B) Seed × fidelity failures
Chaste seed index k
k = 0 5% 10% 25% 30% 35% 35% 25% 50% 55%
k = 1 0% 5% 30% 25% 20% 20% 30% 40% 55%
k = 2 5% 10% 20% 30% 25% 30% 20% 45% 55%
k = 3 5% 15% 35% 40% 35% 35% 35% 35% 45%
k = 4 5% 15% 30% 25% 30% 35% 30% 50% 45%
k = 5 0% 15% 30% 30% 30% 50% 30% 50% 45%
k = 6 0% 10% 40% 35% 40% 40% 40% 35% 55%
k = 7 0% 10% 15% 25% 25% 15% 15% 40% 55%
k = 8 0% 10% 40% 40% 40% 40% 40% 45% 55%
k = 9 0% 10% 20% 30% 35% 35% 20% 50% 55%
T70
T140
T350
ΔT0.02
ΔT0.05
ΔT0.10
K1
K2
K5
Failure rate
(C) Inference methods
HF-SMC-ABC
RESAMPLE & PERTURB
EVALUATE & ACCEPT
UPDATE WEIGHTS & STEP DOWN
HF-NPE
HF-SNPE
MF-SBI
Prior
Observation
Posterior
HF Simulation
LF Simulation
Mixed LF/HF Simulations
Neural Density Estimation
Current Posterior Proposal
Active Acquisition (Uncertainty)